

Write-up: next-season style forecast → AI product concepts (H&M data)

Next-Season Concepts: Autumn 2020

Top-3 forecast winners for 23 Sep–20 Oct 2020 (4-week horizon) → AI-generated next-season concepts

#1 Pluto RW slacks · Trousers

style 0751471



WINNING STYLE



NEW CONCEPT

Critic: approved

Why it won

Rank #1 of 20,318 styles scored (one per garment group): 7,417 units forecast for 23 Sep–20 Oct 2020 (9,182 in the last 4 weeks). Model: +8% vs the last-week × 4 run-rate of 6,844; top upward driver: still near full price, only 2% below its peak price (×1.21).

What changed

Charcoal-and-camel glen-check wool-look fabric; camel contrast side stripe down each leg. Kept: slim tapered ankle-length leg, belt-loop waistband, slant pockets and pressed centre crease.

#2 Lucy blouse · Shirt

style 0762846



WINNING STYLE



NEW CONCEPT

Critic: approved

Why it won

Rank #2 of 20,318 styles scored (one per garment group): 6,492 units forecast for 23 Sep–20 Oct 2020 (6,662 in the last 4 weeks). Model: -5% vs the last-week × 4 run-rate of 6,820; top upward driver: still near full price, only 1% below its peak price (×1.23).

What changed

Deep burgundy satin with a soft sheen; fuller sleeves gathered into deep buttoned cuffs. Kept: shirt collar, open V-neck, button front and rounded hem.

#3 RICHIE HOOD · Hoodie

style 0685814



WINNING STYLE



NEW CONCEPT

Critic: not approved

Why it won

Rank #3 of 20,318 styles scored (one per garment group): 5,486 units forecast for 23 Sep–20 Oct 2020 (6,108 in the last 4 weeks). Model: +18% vs the last-week × 4 run-rate of 4,660; top upward driver: still near full price, only 1% below its peak price (×1.24).

What changed

Heather-grey fabric; twin zip pockets on the pouch; contrast-tipped ribbed cuffs. Kept: drawstring hood and kangaroo pouch. Intended cropped, boxy shape was not achieved by the image model; critic did not approve. The small chest label is still visible.

Final board: 3 forecast winners → new concepts (outputs/final_board.png).

Result: outputs/final_board.png (3 winners → 3 concepts) · **evidence:** outputs/evidence_sheet.png, outputs/evidence/<code>/lineage.json · **agent trace:** outputs/runs/20260928-005623/trace.jsonl.

1. Approach in one paragraph

Transactions (31.8M rows, Sep 2018 – 22 Sep 2020) are aggregated to **styles** (product_code, i.e. all colour variants of one design) per Wed→Tue week. A LightGBM model, trained on 45 weekly snapshots, forecasts each style's units for the next 4 weeks. The top-3 are picked with a diversity rule (one per garment group) and an availability proxy (must have sold in the last 2 weeks). An agentic workflow then turns each winner into a new product concept. Claude Agent SDK orchestrator → forecaster, style-analyst (reusable style-dna-brief skill), designer (FLUX.1 Kontext image edit) and critic (CLIP novelty + visual check) sub-agents, all working through MCP tools. Every step is written to lineage.json.

2. Key choices

- **Style** = product_code. Colourways of one design share the same "DNA"; ranking articles would give three colours of one trouser. Winners sell in 10–23 colours, which is also the evidence that *shape*, not colour, is what customers buy.
- **"Strong"** = most units in the next 4 weeks (buyers are tracked too). Units is what a merchandiser plans.
- **Why a 4-week horizon, and how that relates to "next season"**.
 - *Fast fashion turns over quickly*. The median style sells for 19 active weeks, and 32% of weekly units come from styles launched in the last 12 weeks (EDA). Over a full 12–13-week season, a large share of demand comes from styles that don't exist yet, so a style-level history model can't forecast them. Four weeks is the longest window where "which existing styles will lead" is still well defined and testable.

- *The cutoff is the season boundary.* Data ends 22 Sep 2020, so the 4 weeks (23 Sep – 20 Oct) are the opening of autumn. The seasonal bonus shows the same pipeline switches with the season: at a late-May cutoff swimwear is 38% of the top-100's predicted units, and in late September it's 0%, with upper-body garments up 32 points.
- *"Next season" is carried by the concepts, not the forecast.* The forecast finds the styles winning as the season opens. The generated concepts are next-season *products* that keep those styles' proven DNA, i.e. input to the next buying cycle, not a forecast of it.
- *Honest limit.* The model is not validated beyond 4 weeks. HORIZON_WEEKS is one config value, but a 12-week run was not tested, so no accuracy claim is made for it.
- **Weekly snapshots, no leakage.** Features use only weeks before the cutoff (tests scramble future data and check the features don't change). The target is the 4 weeks from the cutoff.
- **Model target = uplift over the naive run-rate.** A plain log-units regressor lost to "last week × 4" at the top of the ranking, so the model predicts $\log_{1p}(\text{next } 4w) - \log_{1p}(4 \times \text{last week})$, weighted toward high-volume styles. A LambdaRank variant was tried and lost (NDCG@50 0.846 vs 0.944 on the inner cutoffs).
- **Selection:** at most one style per garment group (a board of three trousers is not useful). "Sold in the last 2 weeks" is the *only* availability signal. **The dataset has no stock data** (see §6).

3. Model results: honest version

Validation week (cutoff 26 Aug → 22 Sep 2020, 20,639 styles) and a 10-cutoff rolling backtest (each cutoff scored by a model trained only on earlier data):

	Model	Last week × 4	Last 4 weeks	Same 4 wks last year
NDCG@50, validation week	0.862	0.864	0.569	0.601
precision@12, validation week	0.417	0.500	0.250	0.167
NDCG@50, backtest mean ± std	0.924 ± 0.031	0.906 ± 0.040	0.906 ± 0.016	n/a
Model wins on NDCG@50 (backtest)	n/a	7/10	7/10	n/a

- Against **"last 4 weeks"** (which lags momentum) the win is clear on the validation week (0.862 vs 0.569), but in the 10-week backtest it's the same small margin as against last-week × 4 (0.924 vs 0.906, 7/10 wins). Against the strong naive baseline **"last week × 4" it wins 7/10 backtest weeks on NDCG@50 but only ties it on the validation week**, and on precision@12 it wins 3/10 (6 ties).
- **Where the value is** (outputs/figures/movers.md): when the model ranks a style far *below* the baseline it is usually right (6/10, mostly one-week spikes that fade); when it ranks one far *above* it is usually wrong (3/10, mostly late-summer swimwear). So the model's value is **demoting fading styles, not finding new risers**.
- SHAP (shap_summary.png): one-week spikes get shrunk; heavily discounted, end-of-life styles are faded; young styles get a boost. All three winners share the same top upward driver, *still near full price*.
- **The three winners share a pattern** (checked against forecast.json and the weekly sales): all are early-autumn risers (last 4 weeks = 1.9–2.6× the 4 weeks before) whose September rise the model partly discounts (forecast –19%, –2%, –10% vs the last 4 weeks). They win on sustained volume at near full price (1–2% below peak price), not markdowns.
- Seasonal bonus: at the summer cutoff, 7 of the predicted top-10 were in the actual top-10.

4. From forecast to concept (lineage)

For each winner, outputs/evidence/<code>/ holds forecast.json (rank, forecast, SHAP drivers, attributes) → brief.json (KEEP/CHANGE, validated by skills_lib/style_dna.py) → generation_log.jsonl (exact prompt, seed, model; input photos in outputs/refs/ are H&M/Kaggle data, downloaded by forecasting/select.py and not included) → critic.jsonl (decision, CLIP scores, note) → lineage.json (links all of these, plus the board caption and the briefed changes the image did not render).

Winner	Forecast (next 4 wks)	Final concept	Critic	CLIP to own photos
#1 Pluto RW slacks (trousers)	7,417	concept_2: glen check, camel side stripe	approved	0.726
#2 Lucy blouse	6,492	concept_1: burgundy satin, fuller gathered sleeves	approved	0.690
#3 RICHIE HOOD	5,486	concept_2: grey, zip pockets, striped cuffs	not approved	0.797

Manual interventions (approved by the user, logged with notes): (1) a third RICHIE HOOD attempt beyond the one-revision rule, also rejected, so the board keeps concept_2 and says the critic did not approve it; (2) Pluto regenerated from the run's own brief so its lineage is clean (the run had reused an image made from a hand-written test prompt). Board captions were checked by eye; briefed changes the image model did not render (e.g. Pluto's cropped hem, RICHIE's cropped boxy shape) are not claimed.

5. Agentic workflow

Orchestrator (Claude Agent SDK, Sonnet) with 4 sub-agents, 3 stdio MCP servers (`retail`, `forecast`, `image`) plus an in-process `run` server for deterministic bookkeeping, and 1 skill. Guard-rails live in code, not prompts: a permission gate (whitelist; writes only under `outputs/`), an image budget (≤ 4 new images and ≤ 1 revision per run; identical requests reuse the file), and lineage assembled from files, not typed by the LLM. The run: 7 min, 59 tool calls across 5 agents, 3 GPU generations.

6. Limitations

- **No stock data** → **censored demand**. A style that sold out looks like a weak seller; "sold in the last 2 weeks" only filters out styles that are clearly gone. Real use needs inventory/OTB data.
- **New styles have no history**, and 32% of units come from them. This model can only rank existing styles.
- **2020 is COVID-distorted** (online share 69% → 73%); last-year features are weaker than usual.
- **Image model**: FLUX Kontext changes fabric, colour and trims well but **did not change a garment's proportions** in three hoodie attempts, and small chest labels reappear.
- **CLIP is a weak novelty proxy**: it barely reacts to colour (colourways of one style score 0.80–0.94), so the "too close" threshold was calibrated to 0.80 and the critic also checks the images visually.
- **Brief validator gap**: Pluto's brief KEEPs "solid colour" while CHANGING to a check. The rule check does not catch KEEP-vs-CHANGE contradictions yet.
- Single retailer, single season boundary; the backtest covers 10 summer weeks.

7. Issues we hit (engineering)

- **scikit-learn replaced because of a device policy**. The laptop's Application Control policy blocks sklearn's compiled DLL (`sklearn.utils.murmurhash`). Its only use, `ndcg_score`, is now a pure-numpy function with identical behaviour (tie-averaged gains, log2 discount, 0 when the ideal DCG is 0, mean over rows). It reproduces the sklearn-computed validation `NDCG@50` values exactly, and tests pin hard-coded values, **so all reported results are unchanged**. (shap still imports sklearn, so only the two SHAP *plots* can't be regenerated on that laptop; the SHAP drivers use LightGBM's built-in TreeSHAP.)
- Reproducibility: DuckDB's `mode()` tie-breaking and multithreaded LightGBM made reruns drift; both are fixed.
- Agent SDK: sub-agents run asynchronously in this CLI version, and an unanswerable permission prompt blocked the critic in a dry run. Fixed with a code-based permission gate and a harness that waits for all sub-agents. Four mock dry runs were done before spending real GPU quota.
- Free Hugging Face GPU quota (~5–10 images/day) instead of a paid API: hence the budget guards.

8. What I'd do next at Merchmix

Plug in inventory and open-to-buy so selection is stock-aware; add new-style cold-start (attribute-similarity to past launches); route concepts to a buyer approval queue with the lineage attached; and extend the critic with a shape check (silhouette/edge comparison) instead of relying on CLIP.